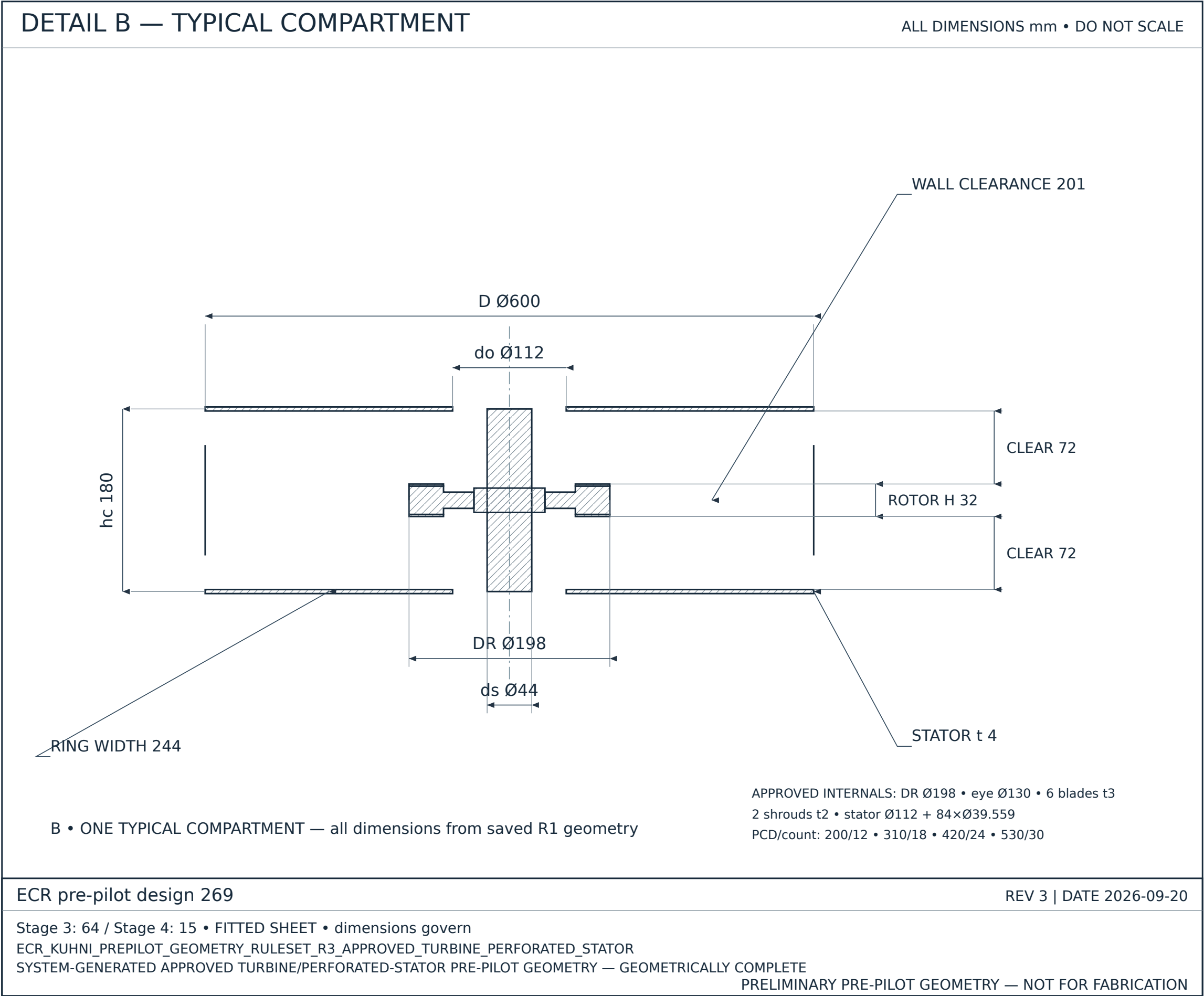
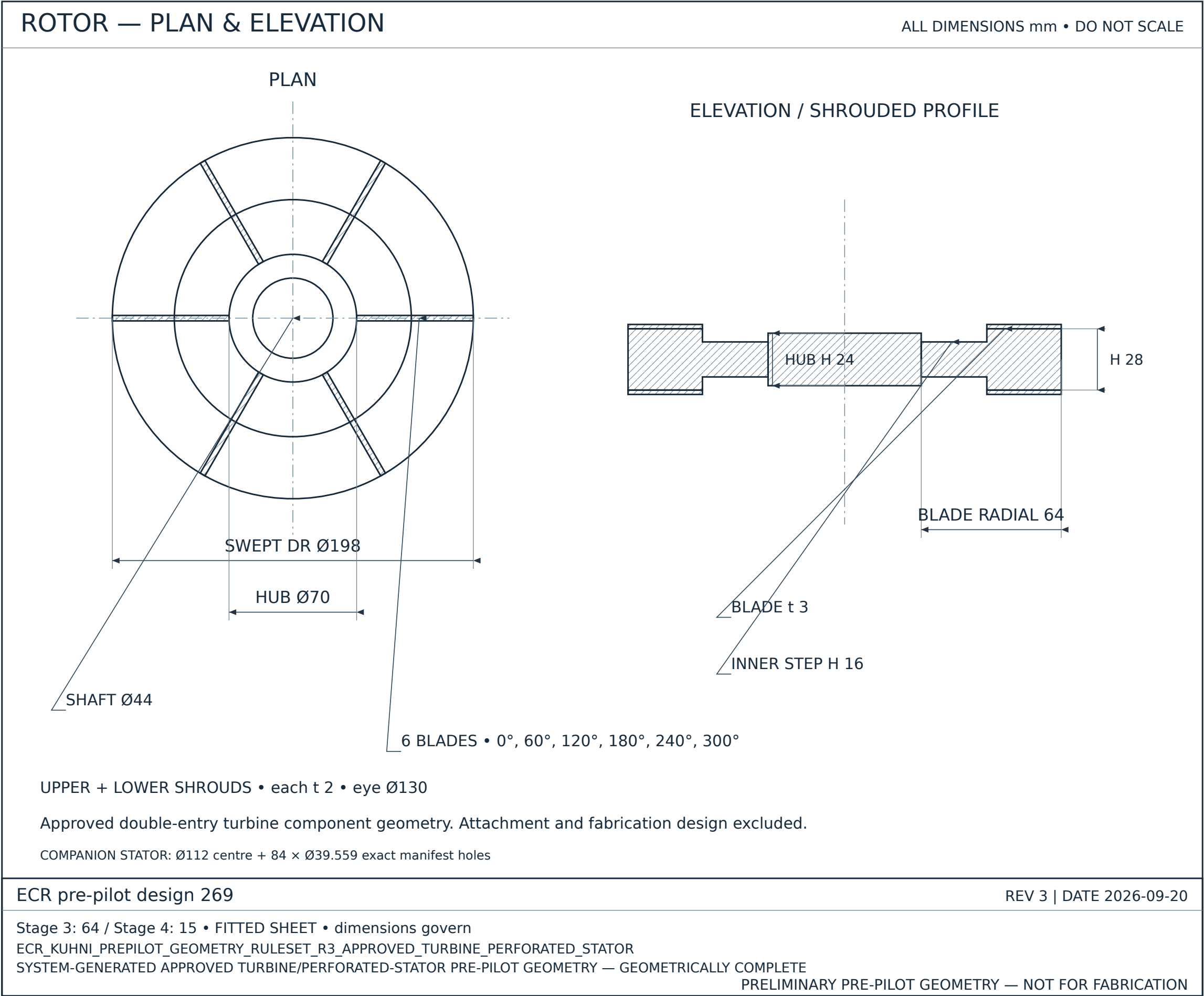
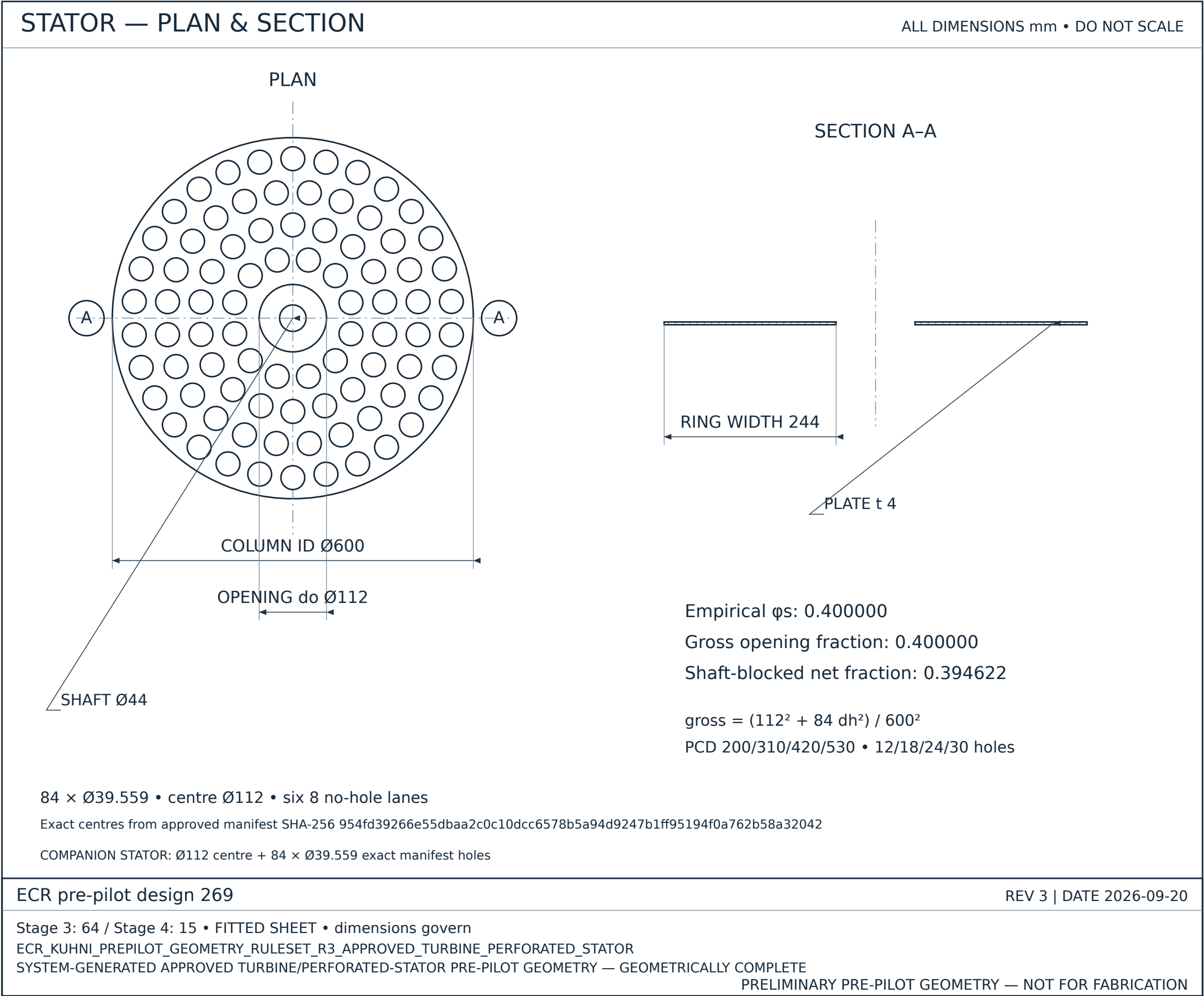








Geometry summary — frozen revision dimensions

PRELIMINARY PRE-PILOT GEOMETRY — NOT FOR FABRICATION

Geometrically complete pre-pilot layout only; not fabrication or mechanical qualification.

sourceHash: 04dfb56f2139741bb5813460c9a575e95b6927718f837ed09629fb70e2bbb610 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

stage3ResultId: 64 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

stage4ResultId: 15 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

columnDiameterM: 0.6 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

compartmentHeightM: 0.18 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rotorDiameterRatio: 0.33 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rotorDiameterM: 0.198 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

statorFreeAreaRatio: 0.4 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

selectedRpm: 45 rpm | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rpmMin: 30 rpm | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rpmMax: 60 rpm | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

phaseConfiguration: nmp-continuous-rrbo-dispersed | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

compartmentCount: 18 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

requiredActiveHeightM: 3.24 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

installedActiveHeightM: 3.24 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

designNt: 7 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

sizingMethod: ADOPTED_COMPARTMENT_EFFICIENCY | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

designCompartmentEfficiency: 0.4 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

impliedInstalledHetsMPerTheoreticalStage: 0.4628571428571429 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rotorOffsetM: 0.09 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. hc/2; centred rotor.

bottomDisengagementM: 0.6 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D; packaging allocation, not separation-time qualification.

topDisengagementM: 0.6 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D; packaging allocation, not separation-time qualification.

bottomHeadDepthM: 0.15 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/4; 2:1 ellipsoidal internal envelope.

topHeadDepthM: 0.15 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/4; 2:1 ellipsoidal internal envelope.

headStraightFlangeM: 0 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Zero in R1 internal envelope model.

supportHeightM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/2; skirt below vessel datum.

skirtDiameterM: 0.6 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D; centred cylindrical envelope.

skirtAccessWidthM: 0.18 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.30D; access opening at azimuth zero.

skirtAccessHeightM: 0.15 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.25D; access opening.

skirtAccessElevationM: -0.15 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. -0.25D; centre elevation.

lowerShaftSupportM: 0.4499999999999996 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. hB+LB/2.

upperShaftSupportM: 4.29 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. z1+LT/2.

supportArmWidthM: 0.012 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.02D; tangential width, three arms at 30/150/270 degrees.

supportArmDepthM: 0.024 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.04D; axial depth.

supportHousingDiameterM: 0.066 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 1.5ds; central envelope.

supportHousingHeightM: 0.024 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.04D; central envelope.

supportArmRadialStartM: 0.033 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.75ds; housing surface.

supportArmRadialEndM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/2; shell interface.

shaftLowerTerminationM: 0.4499999999999996 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Lower support centre.

shaftUpperTerminationM: 4.890000000000001 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. HV+D/4; drive engagement.

shaftLengthM: 4.44 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Upper minus lower termination.

sealElevationM: 4.74 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Top pole; coaxial with shaft.

sealDiameterM: 0.088 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 2ds; reserved housing, not product selection.

sealHeightM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.10D above top pole.

drivePedestalDiameterM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/2; coaxial reserved envelope.

drivePedestalHeightM: 0.15 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/4 above top pole.

driveBodyDiameterM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/2; reserved envelope.

driveBodyHeightM: 0.4499999999999996 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 3D/4 above pedestal.

driveHeightM: 0.6 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D including pedestal; not gearbox adequacy.

columnAreaM2: 0.2827433388230814 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. $\pi D^2/4$.

shaftAreaM2: 0.0015205308443374595 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. $\pi ds^2/4$.

statorGrossOpenAreaM2: 0.11309733552923258 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; recomputed from approved component solids; no upstream feedback.

statorNetOpenAreaM2: 0.11157680468489513 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; recomputed from approved component solids; no upstream feedback.

grossFreeAreaRatio: 0.4000000000000001 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; recomputed from approved component solids; no upstream feedback.

shaftBlockedFreeAreaRatio: 0.394622222222223 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; recomputed from approved component solids; no upstream feedback.

calculatedFreeAreaRatio: 0.4000000000000001 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; recomputed from approved component solids; no upstream feedback.

rotorWallClearanceM: 0.2009999999999998 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; recomputed from approved component solids; no upstream feedback.

shaftOpeningClearanceM: 0.034 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; recomputed from approved component solids; no upstream feedback.

statorRadialWidthM: 0.244 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; recomputed from approved component solids; no upstream feedback.

rotorLowerClearanceM: 0.072 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; recomputed from approved component solids; no upstream feedback.

rotorUpperClearanceM: 0.072 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; recomputed from approved component solids; no upstream feedback.

activeStartM: 0.75 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. hB+LB.

activeEndM: 3.99 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. z0+HA.

topTangentM: 4.59 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. z1+LT.

vesselHeightM: 4.74 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. HA+2.5D.

overallHeightM: 5.640000000000001 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Support+vessel+drive = HA+4D.

flowArrangement: nmp-down-rrho-up | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Explicit application layout; continuity and upstream hydraulics unchanged.

headProfiles: 2:1 ellipsoidal | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Internal envelope only.

bladeAzimuthsDeg: 0, 60, 120, 180, 240, 300 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Six equally spaced blades; same clocking in every compartment.

supportAzimuthsDeg: 30, 150, 270 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Approved three-arm support clocking.

skirtAccessAzimuthDeg: 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Approved skirt access orientation.

designEnvelope: hc/D 0.20–0.30; DR/D 0.33–0.50; ϕ_s 0.20–0.40 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class B. Perry 8e p15-83 → Pratt-Stevens; advisory, never clamp inherited geometry.

P01-elevationM: 0.51 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-boreM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-azimuthDeg: 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-radialOffsetM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-outsideDiameterM: 0.072 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-projectionM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P02-elevationM: 0.39 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-boreM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-azimuthDeg: 180 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-radialOffsetM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-outsideDiameterM: 0.072 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-projectionM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P03-elevationM: 4.23 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P03-boreM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P03-azimuthDeg: 180 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P03-radialOffsetM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

[illegible]

S02-outsideDiameterM: 0.018 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.

S02-projectionM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.

I01-elevationM: 0.3299999999999996 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

I01-boreM: 0.015 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

I01-azimuthDeg: 45 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

I01-radialOffsetM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

I01-outsideDiameterM: 0.018 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

I01-projectionM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

I02-elevationM: 4.17 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

I02-boreM: 0.015 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

I02-azimuthDeg: 45 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

I02-radialOffsetM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

I02-outsideDiameterM: 0.018 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

I02-projectionM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

D01-elevationM: 0 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.

D01-boreM: 0.024 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.

D01-radialOffsetM: 0 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.

D01-outsideDiameterM: 0.0288 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.

D01-projectionM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.

D01-surfaceEdgeElevationMinM: 0 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.

D01-surfaceEdgeElevationMaxM: 0.00017289964762715604 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.

V01-elevationM: 4.71 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

V01-boreM: 0.024 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

V01-azimuthDeg: 315 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

V01-radialOffsetM: 0.18 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

V01-outsideDiameterM: 0.0288 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

V01-projectionM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

V01-surfaceEdgeElevationMinM: 4.704246050259954 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

V01-surfaceEdgeElevationMaxM: 4.715076616519635 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

compartment1-bottomM: 0.75 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment1-topM: 0.9299999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

[illegible]

shaftDiameterM: 0.044 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved shaft diameter.

statorThicknessM: 0.004 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved plate thickness.

bladeCount: 6 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved six-blade architecture.

bladeHeightM: 0.028 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved outer paddle height.

bladeInnerSegmentHeightM: 0.016 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved inner web height.

bladeOuterSegmentHeightM: 0.028 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved outer paddle height.

bladeThicknessM: 0.003 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved blade thickness.

hubDiameterM: 0.07 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved hub diameter.

hubHeightM: 0.024 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved hub height.

rotorAxialEnvelopeM: 0.032 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved total shrouded rotor height.

bladeRadialLengthM: 0.064 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved nominal radial reach from hub OD to rotor OD.

bladeInnerSegmentLengthM: 0.03 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved web radial interval, 35 to 65 mm radius.

bladeOuterSegmentLengthM: 0.034 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved paddle/shroud radial interval, 65 to 99 mm radius.

shroudThicknessM: 0.002 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Each approved upper/lower shroud.

shroudInnerDiameterM: 0.13 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved eye diameter.

shroudOuterDiameterM: 0.198 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved rotor/shroud OD.

statorOpeningDiameterM: 0.112 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved central opening (130 - 2×9).

statorHoleCount: 84 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved exact perforation count.

statorHoleDiameterM: 0.03955947902781812 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved calculated profile-hole diameter.

statorLaneWidthM: 0.008 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Six retained radial no-hole lanes.

rotorConstruction: approved-double-entry-shrouded-turbine | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved component architecture identifier.

statorConstruction: approved-perforated-stator | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR; Class C. Approved perforated plate architecture identifier.

Compartment and internals schedules

Compartment 1: bottom 0.75 m; top 0.9299999999999999 m; rotor 0.84 m; stator 0.9299999999999999 m
Compartment 2: bottom 0.9299999999999999 m; top 1.1099999999999999 m; rotor 1.02 m; stator 1.1099999999999999 m
Compartment 3: bottom 1.1099999999999999 m; top 1.29 m; rotor 1.2 m; stator 1.29 m
Compartment 4: bottom 1.29 m; top 1.47 m; rotor 1.38 m; stator 1.47 m
Compartment 5: bottom 1.47 m; top 1.65 m; rotor 1.56 m; stator 1.65 m
Compartment 6: bottom 1.65 m; top 1.83 m; rotor 1.74 m; stator 1.83 m
Compartment 7: bottom 1.83 m; top 2.01 m; rotor 1.92 m; stator 2.01 m
Compartment 8: bottom 2.01 m; top 2.19 m; rotor 2.0999999999999996 m; stator 2.19 m
Compartment 9: bottom 2.19 m; top 2.37 m; rotor 2.2800000000000002 m; stator 2.37 m
Compartment 10: bottom 2.37 m; top 2.55 m; rotor 2.46 m; stator 2.55 m
Compartment 11: bottom 2.55 m; top 2.73 m; rotor 2.6399999999999997 m; stator 2.73 m
Compartment 12: bottom 2.73 m; top 2.91 m; rotor 2.82 m; stator 2.91 m
Compartment 13: bottom 2.91 m; top 3.09 m; rotor 3 m; stator 3.09 m
Compartment 14: bottom 3.09 m; top 3.27 m; rotor 3.1799999999999997 m; stator 3.27 m
Compartment 15: bottom 3.27 m; top 3.4499999999999997 m; rotor 3.36 m; stator 3.4499999999999997 m
Compartment 16: bottom 3.4499999999999997 m; top 3.63 m; rotor 3.54 m; stator 3.63 m
Compartment 17: bottom 3.63 m; top 3.81 m; rotor 3.7199999999999998 m; stator 3.81 m
Compartment 18: bottom 3.81 m; top 3.9899999999999998 m; rotor 3.9 m; stator 3.9899999999999998 m
R | Approved double-entry radial-flow shrouded six-blade turbine | count 18 | diameter 0.198 m | thickness 0.032 m | elevations 0.84, 1.02, 1.2, 1.38, 1.56, 1.74, 1.92, 2.0999999999999996, 2.2800000000000002, 2.46, 2.6399999999999997, 2.82, 3, 3.1799999999999997, 3.36, 3.54, 3.7199999999999998, 3.9 m
S | Approved perforated stator: central opening plus 84 exact profile holes | count 19 | diameter 0.6 m | thickness 0.004 m | elevations 0.75, 0.9299999999999999, 1.1099999999999999, 1.29, 1.47, 1.65, 1.83, 2.01, 2.19, 2.37, 2.55, 2.73, 2.91, 3.09, 3.27, 3.4499999999999997, 3.63, 3.81, 3.9899999999999998 m
SH | Continuous shaft envelope; mechanical qualification excluded | count 1 | diameter 0.044 m | thickness 0.044 m | elevations 0.4499999999999996, 4.8900000000000001 m
SUP | Three-arm shaft supports, azimuths 30/150/270 deg | count 2 | diameter 0.6 m | thickness 0.024 m | elevations 0.4499999999999996, 4.29 m

Preliminary nozzle / connection schedule

- P01 | RRBO feed | bottom | elevation 0.51 m | bore 0.06 m | azimuth 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.072 m; projection 0.06 m; radial offset 0.3 m
- P02 | NMP-rich extract outlet | bottom | elevation 0.39 m | bore 0.06 m | azimuth 180 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.072 m; projection 0.06 m; radial offset 0.3 m
- P03 | NMP feed | top | elevation 4.23 m | bore 0.06 m | azimuth 180 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.072 m; projection 0.06 m; radial offset 0.3 m
- P04 | RRBO-rich raffinate outlet | top | elevation 4.3500000000000005 m | bore 0.06 m | azimuth 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.072 m; projection 0.06 m; radial offset 0.3 m
- A01 | Low-shell flush | bottom | elevation 0.27 m | bore 0.024 m | azimuth 270 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.0288 m; projection 0.06 m; radial offset 0.3 m
- A02 | High-shell equalization | top | elevation 4.470000000000001 m | bore 0.024 m | azimuth 270 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.0288 m; projection 0.06 m; radial offset 0.3 m
- S01 | Lower sample connection | bottom | elevation 0.63 m | bore 0.015 m | azimuth 90 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.018 m; projection 0.06 m; radial offset 0.3 m
- S02 | Upper sample connection | top | elevation 4.11 m | bore 0.015 m | azimuth 90 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.018 m; projection 0.06 m; radial offset 0.3 m
- I01 | Lower instrument connection | bottom | elevation 0.3299999999999996 m | bore 0.015 m | azimuth 45 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.018 m; projection 0.06 m; radial offset 0.3 m
- I02 | Upper instrument connection | top | elevation 4.17 m | bore 0.015 m | azimuth 45 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.018 m; projection 0.06 m; radial offset 0.3 m
- D01 | Bottom-head drain | bottom-head | elevation 0 m | bore 0.024 m | azimuth N/A axial deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A. | axis down; OD 0.0288 m; projection 0.06 m; radial offset 0 m
- V01 | Top-head crown-region vent | top-head | elevation 4.71 m | bore 0.024 m | azimuth 315 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface z=zT+0.20D; not absolute-high-point or vent-duty qualification. | axis up; OD 0.0288 m; projection 0.06 m; radial offset 0.18 m

Validation, assumptions and unresolved items

- PASS: Frozen sources must be identified, current and compatible.
- PASS: columnDiameterM must be finite and positive.
- PASS: rotorDiameterM must be finite and positive.
- PASS: rotorDiameterRatio must be finite and positive.
- PASS: compartmentHeightM must be finite and positive.
- PASS: compartmentCount must be finite and positive.
- PASS: installedActiveHeightM must be finite and positive.
- PASS: requiredActiveHeightM must be finite and positive.
- PASS: designNt must be finite and positive.
- PASS: selectedRpm must be finite and positive.
- PASS: rpmMin must be finite and positive.
- PASS: rpmMax must be finite and positive.
- PASS: statorFreeAreaRatio must be finite and positive.
- PASS: Positive integer count; drawing capacity 1000 compartments.
- PASS: Frozen count × pitch must equal frozen installed height; no repair.
- PASS: Frozen adopted sizing basis must be Nt=7 and compartment efficiency=0.40.
- PASS: Frozen count must equal ceil(Nt / adopted compartment efficiency).
- PASS: Frozen required and installed heights must both equal count × pitch.
- PASS: Implied installed HETS must equal installed height / Nt and remains diagnostic only.
- PASS: Frozen rotor diameter/ratio must agree and fit column.
- PASS: Frozen selected RPM must lie inside frozen window.
- PASS: Frozen phase continuity is required.
- PASS: Empirical free-area input must be between zero and one.
- PASS: columnDiameterM must be finite.
- PASS: compartmentHeightM must be finite.
- PASS: rotorDiameterRatio must be finite.
- PASS: rotorDiameterM must be finite.
- PASS: statorFreeAreaRatio must be finite.
- PASS: selectedRpm must be finite.
- PASS: rpmMin must be finite.
- PASS: rpmMax must be finite.
- PASS: compartmentCount must be finite.
- PASS: requiredActiveHeightM must be finite.
- PASS: installedActiveHeightM must be finite.
- PASS: designNt must be finite.
- PASS: designCompartmentEfficiency must be finite.
- PASS: impliedInstalledHetsMPerTheoreticalStage must be finite.
- PASS: shaftDiameterM must be finite.
- PASS: statorThicknessM must be finite.
- PASS: bladeCount must be finite.
- PASS: bladeHeightM must be finite.
- PASS: hubDiameterM must be finite.
- PASS: hubHeightM must be finite.
- PASS: bladeThicknessM must be finite.
- PASS: bladeRadialLengthM must be finite.
- PASS: bladeInnerSegmentLengthM must be finite.
- PASS: bladeOuterSegmentLengthM must be finite.
- PASS: bladeInnerSegmentHeightM must be finite.
- PASS: bladeOuterSegmentHeightM must be finite.
- PASS: rotorOffsetM must be finite.

PASS: bottomDisengagementM must be finite.
PASS: topDisengagementM must be finite.
PASS: bottomHeadDepthM must be finite.
PASS: topHeadDepthM must be finite.
PASS: headStraightFlangeM must be finite.
PASS: supportHeightM must be finite.
PASS: skirtDiameterM must be finite.
PASS: skirtAccessWidthM must be finite.
PASS: skirtAccessHeightM must be finite.
PASS: skirtAccessElevationM must be finite.
PASS: lowerShaftSupportM must be finite.
PASS: upperShaftSupportM must be finite.
PASS: supportArmWidthM must be finite.
PASS: supportArmDepthM must be finite.
PASS: supportHousingDiameterM must be finite.
PASS: supportHousingHeightM must be finite.
PASS: supportArmRadialStartM must be finite.
PASS: supportArmRadialEndM must be finite.
PASS: shaftLowerTerminationM must be finite.
PASS: shaftUpperTerminationM must be finite.
PASS: shaftLengthM must be finite.
PASS: sealElevationM must be finite.
PASS: sealDiameterM must be finite.
PASS: sealHeightM must be finite.
PASS: drivePedestalDiameterM must be finite.
PASS: drivePedestalHeightM must be finite.
PASS: driveBodyDiameterM must be finite.
PASS: driveBodyHeightM must be finite.
PASS: driveHeightM must be finite.
PASS: statorOpeningDiameterM must be finite.
PASS: columnAreaM2 must be finite.
PASS: shaftAreaM2 must be finite.
PASS: statorGrossOpenAreaM2 must be finite.
PASS: statorNetOpenAreaM2 must be finite.
PASS: grossFreeAreaRatio must be finite.
PASS: shaftBlockedFreeAreaRatio must be finite.
PASS: calculatedFreeAreaRatio must be finite.
PASS: rotorWallClearanceM must be finite.
PASS: shaftOpeningClearanceM must be finite.
PASS: statorRadialWidthM must be finite.
PASS: rotorAxialEnvelopeM must be finite.
PASS: rotorLowerClearanceM must be finite.
PASS: rotorUpperClearanceM must be finite.
PASS: activeStartM must be finite.
PASS: activeEndM must be finite.
PASS: topTangentM must be finite.
PASS: vesselHeightM must be finite.
PASS: overallHeightM must be finite.
PASS: skirtAccessAzimuthDeg must be finite.
PASS: P01-elevationM must be finite.
PASS: P01-boreM must be finite.
PASS: P01-azimuthDeg must be finite.

PASS: P01-radialOffsetM must be finite.
PASS: P01-outsideDiameterM must be finite.
PASS: P01-projectionM must be finite.
PASS: P02-elevationM must be finite.
PASS: P02-boreM must be finite.
PASS: P02-azimuthDeg must be finite.
PASS: P02-radialOffsetM must be finite.
PASS: P02-outsideDiameterM must be finite.
PASS: P02-projectionM must be finite.
PASS: P03-elevationM must be finite.
PASS: P03-boreM must be finite.
PASS: P03-azimuthDeg must be finite.
PASS: P03-radialOffsetM must be finite.
PASS: P03-outsideDiameterM must be finite.
PASS: P03-projectionM must be finite.
PASS: P04-elevationM must be finite.
PASS: P04-boreM must be finite.
PASS: P04-azimuthDeg must be finite.
PASS: P04-radialOffsetM must be finite.
PASS: P04-outsideDiameterM must be finite.
PASS: P04-projectionM must be finite.
PASS: A01-elevationM must be finite.
PASS: A01-boreM must be finite.
PASS: A01-azimuthDeg must be finite.
PASS: A01-radialOffsetM must be finite.
PASS: A01-outsideDiameterM must be finite.
PASS: A01-projectionM must be finite.
PASS: A02-elevationM must be finite.
PASS: A02-boreM must be finite.
PASS: A02-azimuthDeg must be finite.
PASS: A02-radialOffsetM must be finite.
PASS: A02-outsideDiameterM must be finite.
PASS: A02-projectionM must be finite.
PASS: S01-elevationM must be finite.
PASS: S01-boreM must be finite.
PASS: S01-azimuthDeg must be finite.
PASS: S01-radialOffsetM must be finite.
PASS: S01-outsideDiameterM must be finite.
PASS: S01-projectionM must be finite.
PASS: S02-elevationM must be finite.
PASS: S02-boreM must be finite.
PASS: S02-azimuthDeg must be finite.
PASS: S02-radialOffsetM must be finite.
PASS: S02-outsideDiameterM must be finite.
PASS: S02-projectionM must be finite.
PASS: I01-elevationM must be finite.
PASS: I01-boreM must be finite.
PASS: I01-azimuthDeg must be finite.
PASS: I01-radialOffsetM must be finite.
PASS: I01-outsideDiameterM must be finite.
PASS: I01-projectionM must be finite.
PASS: I02-elevationM must be finite.

PASS: I02-boreM must be finite.
PASS: I02-azimuthDeg must be finite.
PASS: I02-radialOffsetM must be finite.
PASS: I02-outsideDiameterM must be finite.
PASS: I02-projectionM must be finite.
PASS: D01-elevationM must be finite.
PASS: D01-boreM must be finite.
PASS: D01-radialOffsetM must be finite.
PASS: D01-outsideDiameterM must be finite.
PASS: D01-projectionM must be finite.
PASS: D01-surfaceEdgeElevationMinM must be finite.
PASS: D01-surfaceEdgeElevationMaxM must be finite.
PASS: Entire neck cylinder fits ellipsoidal head.
PASS: V01-elevationM must be finite.
PASS: V01-boreM must be finite.
PASS: V01-azimuthDeg must be finite.
PASS: V01-radialOffsetM must be finite.
PASS: V01-outsideDiameterM must be finite.
PASS: V01-projectionM must be finite.
PASS: V01-surfaceEdgeElevationMinM must be finite.
PASS: V01-surfaceEdgeElevationMaxM must be finite.
PASS: compartment1-bottomM must be finite.
PASS: compartment1-topM must be finite.
PASS: compartment1-rotorM must be finite.
PASS: compartment2-bottomM must be finite.
PASS: compartment2-topM must be finite.
PASS: compartment2-rotorM must be finite.
PASS: compartment3-bottomM must be finite.
PASS: compartment3-topM must be finite.
PASS: compartment3-rotorM must be finite.
PASS: compartment4-bottomM must be finite.
PASS: compartment4-topM must be finite.
PASS: compartment4-rotorM must be finite.
PASS: compartment5-bottomM must be finite.
PASS: compartment5-topM must be finite.
PASS: compartment5-rotorM must be finite.
PASS: compartment6-bottomM must be finite.
PASS: compartment6-topM must be finite.
PASS: compartment6-rotorM must be finite.
PASS: compartment7-bottomM must be finite.
PASS: compartment7-topM must be finite.
PASS: compartment7-rotorM must be finite.
PASS: compartment8-bottomM must be finite.
PASS: compartment8-topM must be finite.
PASS: compartment8-rotorM must be finite.
PASS: compartment9-bottomM must be finite.
PASS: compartment9-topM must be finite.
PASS: compartment9-rotorM must be finite.
PASS: compartment10-bottomM must be finite.
PASS: compartment10-topM must be finite.
PASS: compartment10-rotorM must be finite.
PASS: compartment11-bottomM must be finite.

[illegible]

OUTSIDE ENGINEERING SCOPE: Separation residence-time, connection hydraulic capacity and operational venting qualification
Engineering notes: ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; system-generated; no normal-user construction inputs.
Revision notes: Approved frozen shrouded turbine and 84-hole perforated stator integrated; inherited 18-compartment / 3240-mm active stack. Historical revisions preserved. Preliminary, not for fabrication.
Stage 3 result: 64; Stage 4 result: 15
Source SHA-256: 04dfb56f2139741bb5813460c9a575e95b6927718f837ed09629fb70e2bbb610
PRELIMINARY — NOT FOR FABRICATION. Mechanical verification, pressure-vessel design, shaft strength/deflection/critical speed, bearing/seal selection, drive adequacy and structural supports are outside this package.